MAC21 — Last-Minute Prediction Sheet

The questions most likely to appear, by unit, with the exact section they sit in. **Answers are given** where the question has repeated with the same numbers — memorise those outright. Every value below was computed and checked.

> **Two different kinds of repeat — do not confuse them.**
> **EXACT n×** the *same question with the same numbers* appeared n times. The answer in the last column is the real answer. **Memorise it.**
> **TOPIC n×** the same *technique* appeared n times, but with **different numbers every time**. The answer shown is from one specific paper — it is there as a worked example, **not** as something to reproduce. Learn the method.
> The red count (**4×/6**) is out of the papers carrying that unit (Units I–III: 6 papers · IV: 5 · V: 3). Nothing is guaranteed — but if you read one page, read this one.

UNIT I — Q1 or Q2

QUESTION	SECTION	MK	SEEN	ANSWER / WHAT TO WRITE
Expand $xy + \cos(xy)$ **about** $(1, \pi/2)$, 2nd degree	Q2(c)	7	4×/6 EXACT 2×	**Answer = π/2 — every derivative vanishes at that point.** Show the table of f, f_x, f_y, f_{xx}, f_{xy}, f_{yy} all equal 0, then state f ≈ π/2. Do not panic and hunt for an error. *$xy+\cos(xy)$ is exact in 2 papers; 2 more use a variant — but **the point (1, π/2) never changes.*** *sin variant:* $f=1+\pi/2$, $f_x=\pi/2$, $f_y=1$, $f_{xx}=-\pi^2/4$, $f_{xy}=1-\pi/2$, $f_{yy}=-1$ *xy^2 variant:* $f=\pi^2/4$, $f_x=\pi(\pi-2)/4$, $f_y=\pi-1$, $f_{xx}=0$, $f_{xy}=\pi-1$, $f_{yy}=2$
Newton-Raphson system $x^2+y^2=x$, $x^2-y^2=y$, from $(0.8, 0.4)$, 2 iterations	Q1(c)	7	4×/6 START VARIES	$J = [[2x-1, 2y],[2x, -2y-1]]$ **The equations are identical every time — the starting point is NOT.** All four answers: $(0.8, 0.4)$ 2 iter → **0.771846, 0.419644** $(0.8, 0.3)$ 2 iter → **0.772041, 0.419921** $(0.8, 0.4)$ 1 iter → **0.772881, 0.420339** $(1.0, 0.5)$ 2 iter → **0.773720, 0.420557** *Read the start point off the paper. f = 0 at (0.8,0.4) is not an error — g ≠ 0.*
Show $xy(1-x-y)$ is max at $(1/3, 1/3)$	Q2(b)	4	3×/6 EXACT 3×	r-s-t test. At (1/3,1/3): $rt-s^2 > 0$ and $r < 0$ → **maximum**
Open-top box, volume $108\ ft^3$, minimum surface area	Q1(d)/Q2(d)	7	2×/6 EXACT 2×	Minimise $xy+2xz+2yz$ s.t. $xyz=108$ → **x=6, y=6, z=3, S=108** *Open top = ONE xy face, not two.*
Negative root of $xe_x - \sin x = 0$, 3 decimals	Q2(b)/(c)	4–7	2×/6 EXACT 2×	**Bracket first:** f(−2.9)=+0.08, f(−3.0)=−0.008 → start at −3 **root = −2.991** *x=0 is also a root — starting near 0 collapses to it and fails.*
Lagrange multipliers — constrained optimisation word problem	Q1(d)/Q2(d)	7	4×/6 TOPIC	**A different problem every time** — hottest point on a sphere, open-top box, max/min distance to a sphere, divide 24 into three parts. Same method: $F = f - \lambda g$, solve $F_x=F_y=F_\beta=0$ with $g=0$, divide equations to kill λ early. *Sphere example:* $x^2=y^2=1/4,\ z^2=1/2$ → *T = 50*

2-markers (pick the easier of the two): geometrical interpretation of Newton-Raphson · define algebraic & transcendental equations · state Maclaurin's series (one variable) · state Taylor's series (two variables) · define maxima and minima of two variables

UNIT II — Q3 or Q4

QUESTION	SECTION	MK	SEEN	ANSWER / WHAT TO WRITE
Runge-Kutta 4th order	Q3(d)/Q4(d)	7	6×/6 TOPIC	**In every paper.** $k_1=hf(x,y)$, $k_2=hf(x+h/2,\ y+k_1/2)$, $k_3=hf(x+h/2,\ y+k_2/2)$, $k_4=hf(x+h,\ y+k_3)$, $\Delta y=(k_1+2k_2+2k_3+k_4)/6$ *P26 example:* $y'=y-x^2,\ y(2)=5,\ h=0.1$ → **y(2.1)=5.084146**
Modified Euler's method, 2 corrector iterations	Q4(c)	7	5×/6 TOPIC	**Sits at 4(c) in 4 of 5 papers.** Predictor $y+hf$, then $y_n + (h/2)[f(x_n,y_n) + f(x_{-1}, y^k)]$ repeatedly. *P26:* $y'=-y/(1+x),\ y(3)=2,\ h=0.2$ → **y(3.2)=1.9047**
Newton's law of cooling — coffee 92°C, room 24°C, 80°C in 1 min, when 65°C?	Q3(d)/Q4(d)	7	5×/6 topic EXACT 2×	$T = 24 + 68e^{-kt}$; $k = \ln(17/14) = 0.194156$ **t = 2.606 min** *92/24/80/65 is exact in 2 papers. The other three coolings differ: M24 uses 90/24/75/60, M23 an object 300→250°F, MAM24 a substance 100→70°C. Same working throughout.*

Taylor's series method $dy/dt = e^{-2t} - 2y$, $y(0)=0.1$, 4th degree	Q3(c)	7	2×/6 EXACT 2×	$y'=0.8$, $y''=-3.6$, $y'''=11.2$, $y^{iv}=-30.4$ **$y(0.1) = 0.163740$**
Show $x^2/(a^2+\lambda) + y^2/(b^2+\lambda) = 1$ is self-orthogonal	Q4(c)/(d)	7	2×/6 EXACT 2×	Eliminate λ, then show the DE is **unchanged** when $y' \to -1/y'$. **That invariance IS the proof** — do not solve for λ.
Bungee jumper, $m=68.1$, $c=0.25$, Euler, 1-second steps	Q4(b)	4	2×/6 EXACT 2×	$c/m = 0.003671$; v: 0 → 9.80 → 19.25 → **27.69 m/s**
Orthogonal trajectories (Cartesian or polar)	Q3(b)/(d)	4–7	6×/6 TOPIC	**Eliminate the parameter FIRST**, then swap $y' \to -1/y'$ (polar: $dr/d\theta \to -r^2 d\theta/dr$), then solve.
2-markers: DE of closed L–C circuit without e.m.f. $L\,q'' + q/C = 0$ · write Euler's formula · define orthogonal trajectories · steps to find orthogonal trajectories · write RK4 formula · two differences between analytical and numerical methods				

UNIT III — Q5 or Q6 — your best unit, answer Q5

QUESTION	SECTION	MK	SEEN	ANSWER / WHAT TO WRITE
Prove $[1/(D+a)]X = e^{-ax}\int Xe^{ax}dx$	Q5(b)	4	4×/6 EXACT 4×	**Memorise this outright — it never changes.** Let $y = [1/(D+a)]X$, so $(D+a)y = X$, i.e. $y' + ay = X$. Integrating factor e^{ax} gives $(ye^{ax})' = Xe^{ax}$, so **$y = e^{-ax}\int Xe^{ax}dx$**
Variation of parameters	Q5(c)/(d)	7	5×/5 TOPIC	**In every paper carrying this unit.** $W = y_1y_2' - y_2y_1'$; $y_p = -y_1\int(y_2R/W) + y_2\int(y_1R/W)$ *Standard form first — coefficient of y″ must be 1.* RHS is always awkward ($\sec^3 x$, $\tan x$, e^{3x}/x^2) — that is the signal.
Cauchy's equation $x^2y'' + \dots$	Q5(c)/(d)	7	4×/6 TOPIC	Put $x = e^z$: $xy' = Dy$, **$x^2y'' = D(D-1)y$**, $x^3y''' = D(D-1)(D-2)y$. Solve as constant-coefficient in z, then put $z = \log x$. *The $-D$ is the most-missed detail in the unit.*
Legendre's equation $(ax+b)^2y'' + \dots$	Q6(c)/(d)	4–7	4×/6 TOPIC	Same as Cauchy with $ax+b = e^z$ and extra factors: $(ax+b)y' = aDy$, **$(ax+b)^2y'' = a^2D(D-1)y$** *Forgetting a^2 wrecks every root.*
CF + P.I. for constant coefficients	Q5(c)/Q6(c)/(d)	7	6×/6 TOPIC	Repeated root m (k times) → $(C_1+C_2x+\dots)e^{mx}$; complex $\alpha\pm i\beta$ → $e^{\alpha x}(C_1\cos\beta x + C_2\sin\beta x)$ P.I.: $e^{ax} \to 1/f(a)$ · $\sin/\cos \to D^2 \to -a^2$ · $e^{ax}V \to e^{ax}[1/f(D+a)]V$ · x^n → expand in *ascending* powers of D
2-markers: steps for solving Cauchy's LDE Q5(a) · define linear and non-linear DE with example Q6(a) 3× · define boundary value problem · write the CF given the roots of the A.E. · define non-homogeneous 2nd-order LDE				

UNIT IV — Q7 or Q8

QUESTION	SECTION	MK	SEEN	ANSWER / WHAT TO WRITE
Newton's divided difference — cubic through $(2,4)(4,56)(9,711)(10,980)$, find $y(1.5)$	Q7(d)/Q8(c)	7	5×/5 topic EXACT 2×	Differences: 1st $26, 131, 269$; 2nd $15, 23$; 3rd 1 **$P(x) = x^3 - 2x$** **$y(1.5) = 0.375$** *Divided difference as a technique is in all 5 papers; **this exact data set is in 2**. If your expansion has stray x² or constant terms, you slipped.*
Missing terms — $x = 45,50,55,60,65$; $y = 3.0, -, 2.0, -, -2.4$	Q8(b)	4	3×/5 topic EXACT 2×	3 known values → quadratic → set **both** 3rd differences to zero: $3a+b-9=0$ and $-a-3b+3.6=0$ → **$y(50)=2.925$, $y(60)=0.225$** *Check: 2nd differences all −0.85. One equation is not enough. **This exact table is in 2 papers**; M24 asked the same question with different data.*
Grouped-frequency estimation (income band / age band / marks band)	Q7(c)/Q8(c)	7	3×/5 TOPIC	**Build the CUMULATIVE ("less than") column first** — that is the whole trick. Interpolate on the cumulative values, then **subtract** to get the band count. *P26: marks 70–75 → y(75)=172.80, minus 159 → **≈ 14 students***
Simpson / trapezoidal on $\int_0^1 dx/(1+x^2)$	Q7(d)/Q8(b)/(d)	4–7	3×/5 SAME INTEGRAL	Exact $= \pi/4 = 0.785398$. Ordinates at $h=1/6$: $1, 0.972973, 0.9, 0.8, 0.692308, 0.590164, 0.5$ Trapezoidal (4 strips) **0.782794** · Simpson 1/3 **0.785398** · Simpson 3/8 **0.785396** → ×4 gives **$\pi = 3.141592$** *"Seven ordinates" = 6 intervals. 1/3 needs even n; 3/8 needs n divisible by 3.*

Lagrange interpolation (or inverse)	Q7(b)/(c)	4–7	3×/5 TOPIC	$y = \Sigma\, y_i\, \Pi_{j\neq i}(x-x_j)/(x_i-x_j)$. Inverse = swap x and y roles. *Example: (−1,9)(0,5)(2,3) →* ***P(x)=x²−3x+5***, *f(1)=3*
2-markers: Lagrange's inverse interpolation formula Q8(a) · express $\Delta^2 y_n$ / $\nabla^2 y_n$ in terms of y Q7(a) · state Simpson's 1/3 rule · formula for y', y'' from Newton-Gregory forward · define forward and backward difference				

UNIT V — Q9 or Q10 — only 3 papers exist, so 3× = a clean sweep

QUESTION	SECTION	MK	SEEN	ANSWER / WHAT TO WRITE
System of ODEs by matrix method	Q10(d)	7	3×/3 TOPIC	**The final question of the exam in all three papers.** Eigenvalues of A → eigenvectors → $X = c_1 V_1 e^{E_1 t} + c_2 V_2 e^{E_2 t}$ → apply initial conditions. *Rabbit/wolf: λ=2,3 ;* ***r = 360e²ᵗ − 120e³ᵗ***, ***w = 360e²ᵗ − 60e³ᵗ***
Gauss-Seidel — x+8y+2z=−4 ; 10x+2y+z=59 ; 2x−y+20z=74	Q10(b)/(c)	4–7	3×/3 topic EXACT 2×	**REORDER FIRST** — as printed it is not diagonally dominant. Put the 10x row first, then the 8y row, then the 20z row. It1 (5.9, −1.2375, 3.048125) · It2 (5.842688, −1.992367, 3.016113) Converges to **x=6, y=−2, z=3** *This exact system appears in the model paper and the 2026 paper. M23 and M24 used different systems — so drill the method, and take the answer as a bonus if the numbers match.*
Diagonalization and hence A^n	Q9(c)/(d)	7	3×/3 TOPIC	P = eigenvectors as columns, D = eigenvalues in the **same order**; $A^n = PD^nP^{-1}$ *A=[[1,−1],[2,4]] : λ=2,3 , P=[[−1,−1],[1,2]] ,* ***A⁸ = [[−6049,−6305],[12610,12866]]***
Rayleigh's power method	Q9(c)/(d)	4–7	3×/3 TOPIC	Y = AX, factor out the largest component, repeat. **TRAP:** on [[2,−1,0],[−1,2,−1],[0,−1,2]] from $[1,1,1]^T$ the first iterates go 1, 2, −4 — wild. It needs ~10 iterations to reach **λ = 2+√2 = 3.4142**. If it is still jumping, keep going and say so.
Gauss elimination word problem (dietician / electrical network)	Q9(d)/Q10(c)	7	2×/3 TOPIC	Write the system from the table by **row** (one nutrient = one equation), then row-reduce. *Dietician:* ***x=0.625, y=3.125, z=5 grams***
Rank / consistency of AX=B	Q9(b)/Q10(c)	4–7	2×/3 TOPIC	rank(A)≠rank(A|B) → inconsistent · =n → unique · =r<n → n−r free parameters. *For x+2y−z=a; 2x+y+z=b; x−y+2z=c the condition is* ***b = a+c***
2-markers: define rank · define row echelon form · consistency conditions for AX=B · define modal and spectral matrix · is every square matrix diagonalizable? (**no** — needs n independent eigenvectors) · geometrical interpretation of infinitely many solutions (the two lines coincide)				

In the hall

1. Read all ten questions (5 min). In each unit pick the side where you can start *more parts*.
2. All five 2-markers first — 10 marks in 15 minutes while you are calm.
3. All five 4-markers — you are at 30 marks by minute 50.
4. Then the 7-markers, easiest unit first, ~24 min each.
5. Never leave a part blank. Formula + setup + first two steps is routinely 3 of 7 for ninety seconds. Five partial attempts beat two perfect answers and three blanks.
6. Show every table. These are method-marked — a Gauss-Seidel or RK4 table with a slip in the last row still scores most of the marks.

Compiled from 6 papers plus the official model paper and course handbook. Repeat counts are out of the papers carrying each unit (I–III: 6 · IV: 5 · V: 3). All numerical answers were computed and verified. Predictions are inferences from observed frequency and slot position, not guarantees — read both options before committing.